Problem Set 8

Problem 5.2

A Carnot engine receives 250 kJ s^{-1} of heat from a heat source reservoir at 525°C and rejects heat to a heat-sink reservoir at 50°C. What are the power developed and the heat ejected?

Problem 5.6

Which is the more effective way to increase the thermal efficiency of a Carnot engine: to increase T_H with T_C constant, or to decrease T_C with T_H constant? For a real engine, which would be the more practical way?

Problem 5.8

With respect to 1 kg of liquid water:

(a) Initially at 0°C, it is heated to 100°C by contact with a heat reservoir at 100°C. What is the entropy change of the water? What is the entropy change of the heat reservoir? What is ΔS_{total}?

(b) Initially at 0°C, it is first heated to 50°C by contact with a heat reservoir at 50 °C, and then heated to 100°C by contact with a heat reservoir at 100°C. What is ΔS_{total}?

(c) Explain how the water might be heated from 0 to 100°C so that $\Delta S_{total} = 0$.

Problem 5.10

An ideal gas, $C_P = (7/2)R$, is heated in a steady-flow heat exchanger from 70 to 190°C by another stream of the same gas, which enters at 320°C. The flow rates of the two streams are the same, and heat losses from the exchanger are negligible.

(a) Calculate the molar entropy changes of the two gas streams for both parallel and countercurrent flow in the exchanger.
(b) What is ΔS_{total} in each case?
(c) Repeat (a) and (b) for countercurrent flow if the heating stream enters at 200°C.